LeSuccess Academy

Learn . Educate . Succeed | Official Course Syllabus

Artificial Intelligence and Machine Learning

Duration: 3-6 Months | Mode: HYBRID | 100% Placement Support

Detailed Curriculum & Module Outline:

Module 1: Python and Maths for AI

Python, NumPy, and the linear algebra, probability and calculus that training actually uses

Module 2: Data Preparation and Feature Engineering

Cleaning, encoding, scaling and splitting data without leaking the answer into training

Module 3: Supervised Learning

Regression, classification, decision trees and ensembles with Scikit-learn, on real datasets

Module 4: Unsupervised Learning

Clustering, dimensionality reduction and anomaly detection, and how to judge them without label

Module 5: Model Evaluation and Optimisation

Cross-validation, confusion matrices, ROC curves, regularisation and hyperparameter tuning

Module 6: Neural Network Fundamentals

Perceptrons, activation functions, backpropagation and gradient descent, built from scratch onc

Module 7: Deep Learning with TensorFlow and Keras

Building, training and checkpointing networks, and reading a training curve that has gone wrong

Module 8: PyTorch Essentials

Tensors, autograd and custom training loops - the framework most research code you will read us

Module 9: Computer Vision with CNNs

Convolutions, pooling, augmentation and transfer learning on an image classification project

Module 10: Natural Language Processing

Text preprocessing, embeddings, sequence models and an introduction to transformer architecture

Module 11: Model Deployment and MLOps Basics

Serving a model behind an API, versioning, monitoring and retraining as data drifts

Module 12: Capstone AI Project

A complete project from dataset to deployed model, defended in a technical review